EXAMPLE 12

Pounding of Two Planar Frames, Nonlinear Time History Analysis

Problem Description

A two-bay, seven-story plane frame is linked to a one-bay four-story plane frame using ETABS GAP elements. The structure experiences pounding because of ground motion. The El Centro 1940 (N-S) record is used in the nonlinear time history analysis.

This example illustrates the use of gap elements to model pounding between buildings.

Geometry, Properties and Loads

The geometry of the structure is shown in Figure 12-1.

The combined structure is modeled as a single frame with five column lines and three beam bays. Kip-inch-second units are used. The modulus of elasticity is taken as 29500 ksi. Column and beam section properties are user defined.

Through the joint assignment option, Column lines 4 and 5 are connected to Diaphragm 2. Column lines 1 to 3 remain connected to Diaphragm 1 by default. This arrangement physically divides the structure into two parts. The interaction is provided via the gap elements, which are used as links spanning Column lines 3 and 4. The local axis 1 of the links is in the global X-direction.

Technical Features of ETABS Tested

- Two-dimensional frame analysis
- Use of uniaxial gap elements
- Point assignments
- Nonlinear time history analysis
- Use of multiple diaphragms

Results Comparison

The example frame analyzed using ETABS is also analyzed using SAP2000 (Computers and Structures 2002) for time history loads (SAP2000 has been verified independently). A comparison of key ETABS results with SAP2000 is presented in Table 12-1.

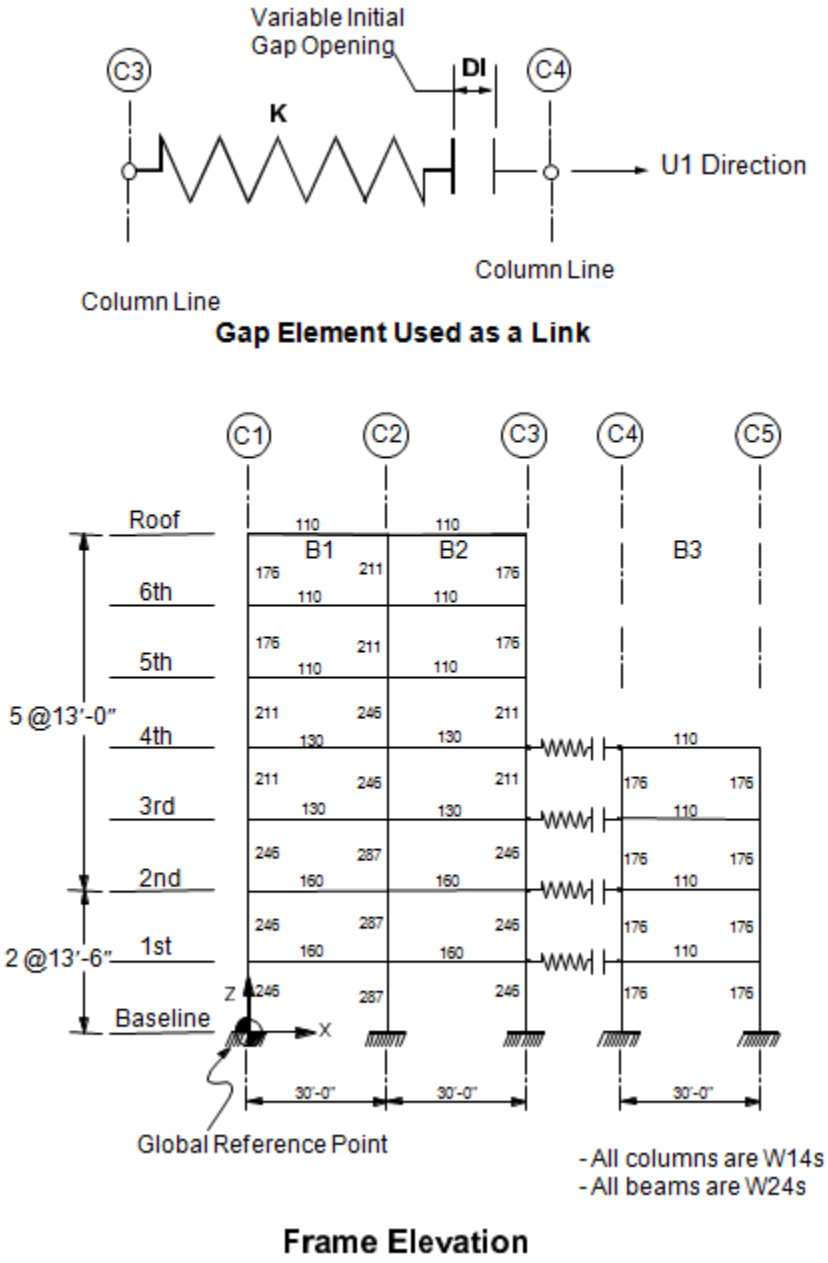


Figure 12-1 Planar Frame with Gap Elements

Table 12-1 Comparison of Results for Time History Analysis

Quantity	ETABS	SAP2000
Maximum Lateral Displacement at Roof	5.5521	5.5521
Maximum Axial Force, Column C1 at ground	266.89	266.88

A typical output produced by the program is shown in Figure 12-2. It shows the variations of the displacement of Column lines 3 and 4 and the link force at Story 4. It is clearly evident that the link force is generated whenever the two column lines move in phase and their separation is less than the specified initial opening or if they move towards each other out of phase. For display purposes, the link forces are scaled down by a factor of 0.01.

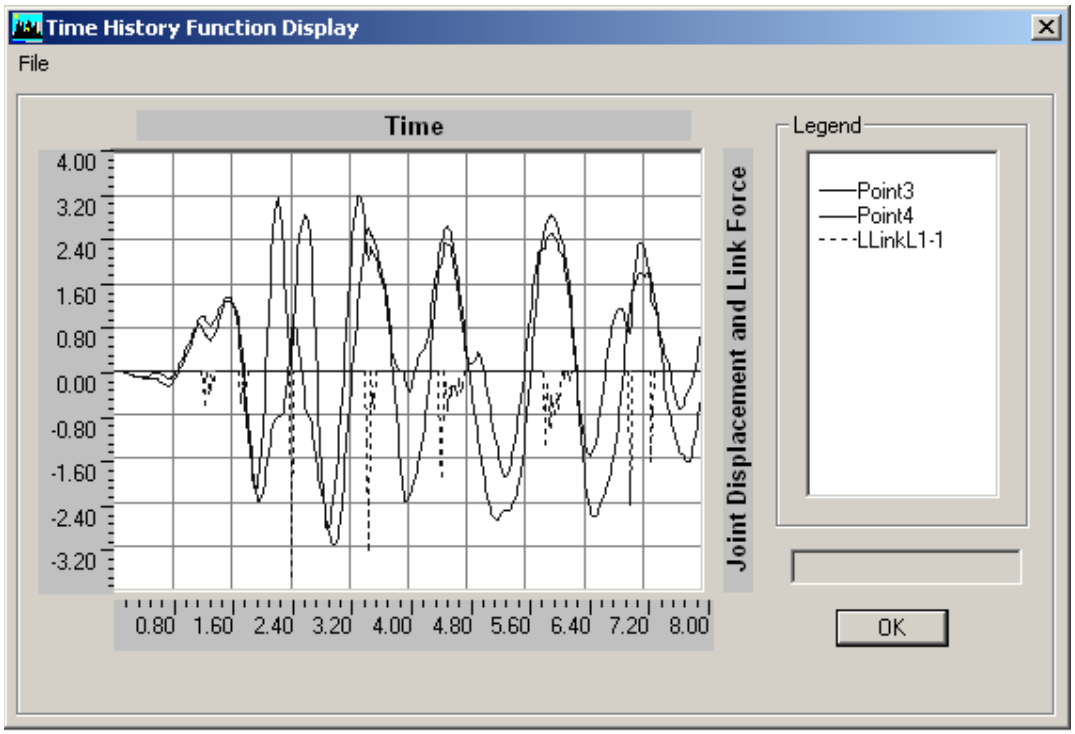


Figure 12-2 Variations of Displacement of Column Lines 3 and 4 and Link Force at Story 4

Computer Files

The input data for this example is Example 12.EDB. The time history file is ELCN-THU. Both of the files are provided as part of the ETABS installation.

Conclusions

The results comparison shows essentially exact agreement between ETABS and SAP2000.